COMPUTATIONAL BEAM ANALYSIS

Compact formula reference • Euler–Bernoulli beam theory • Static downward loading

Key Features

Graphical user interface	Interactive desktop application built with PySide6.
Analytical beam solver	Calculates support reactions, maximum shear force, bending moment, and deflection for the implemented beam cases.
Response visualization	Plots deflected shape, Shear Force Diagram (SFD), and Bending Moment Diagram (BMD).
JSON input save/load	Save beam input data to JSON files and load previously saved inputs to reuse beam configurations.
Input validation	Checks numerical inputs and supported loading configurations.
Calculation validation	Representative analytical cases are checked against independent manual calculations.

The original formula reference follows on the next page and is retained unchanged.

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Notation and units

Symbol	Meaning	Input unit
L	Beam span	m
P	Point-load magnitude	kN
w	Full-span UDL intensity	kN/m
a	Distance from A/fixed end to point load	m
b = L – a	Distance from load to right support	m
E, I	Young's modulus; second moment of area	MPa (N/mm²); mm■

Output units: reactions and shear = kN; moment = kN·m; deflection = mm. Use consistent units in equations: convert L and a to mm and P to N when E is N/mm² and I is mm■. Numerically, 1 kN/m = 1 N/mm.

1. Simply supported — point load P at distance a

Reactions: $R_A = Pb/L$; $R_B = Pa/L$

Maximum shear magnitude: $V_{max} = \max(R_A, R_B)$

Maximum moment: $M_{max} = Pab/L$ at $x = a$

Let $c = \min(a, b)$. **Maximum deflection magnitude:** $\delta_{max} = Pc(L^2 - c^2)^{3/2} / (9\sqrt{3}\,LEI)$

Location of maximum deflection from A: if $a \le b$: $x_{\delta,max} = L - \sqrt{[(L^2 - a^2)/3]}$; otherwise $x_{\delta,max} = \sqrt{[(L^2 - b^2)/3]}$.

Central load (a = b = L/2): $R_A = R_B = P/2$; $M_{max} = PL/4$; $\delta_{max} = PL^3/(48EI)$.

2. Simply supported — full-span UDL w

Reactions: $R_A = R_B = wL/2$

Maximum shear magnitude: $V_{max} = wL/2$

Maximum moment: $M_{max} = wL^2/8$ at $x = L/2$

Maximum deflection: $\delta_{max} = 5wL■/(384EI)$ at $x = L/2$

3. Cantilever — point load P at distance a from fixed end

Fixed support: $R_A = P$; $|M_A| = Pa$

Maximum shear magnitude: $V_{max} = P$

Maximum moment magnitude: $M_{max} = Pa$ at the fixed support

Maximum deflection at free end: $\delta_{max} = Pa^2(3L - a)/(6EI)$, for $0 < a \le L$

End load (a = L): $\delta_{max} = PL^3/(3EI)$

4. Cantilever — full-span UDL w

Fixed support: $R_A = wL$; $|M_A| = wL^2/2$

Maximum shear magnitude: $V_{max} = wL$

Maximum moment magnitude: $M_{max} = wL^2/2$ at the fixed support

Maximum deflection at free end: $\delta_{max} = wL■/(8EI)$

Assumptions and scope

• Prismatic beam with constant EI; linear-elastic material behaviour; small deflections and rotations.

• Idealized supports, static downward vertical loads, and Euler–Bernoulli theory (shear deformation neglected).

• Point load acts within the span; UDL covers the full span. Maximum values are magnitudes; diagram signs follow the solver convention.

• Only implemented and verified cases should be described as supported. Fixed-Fixed and Propped Cantilever remain unsupported unless their solvers are complete.

• This is not a design-code compliance check. Independently verify equations, units, inputs, and all applicable design requirements before real-world use.

Project stack: Python • PySide6 • NumPy • Matplotlib • JSON